mac2texpdf3 Software Documentation

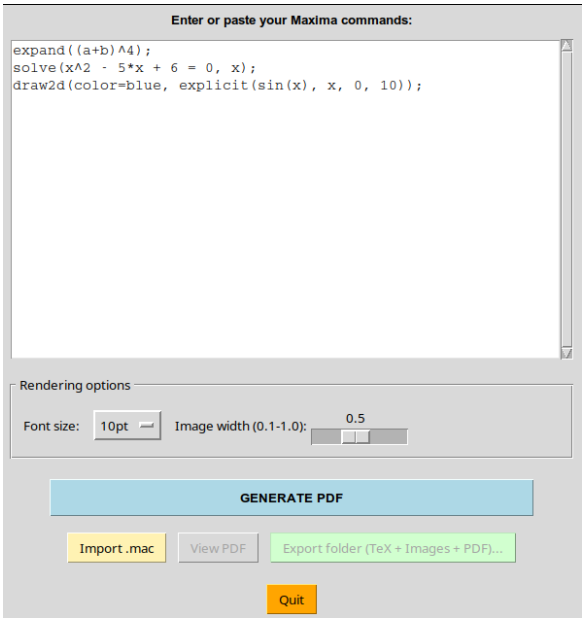
Version 3.0 – October 23, 2026

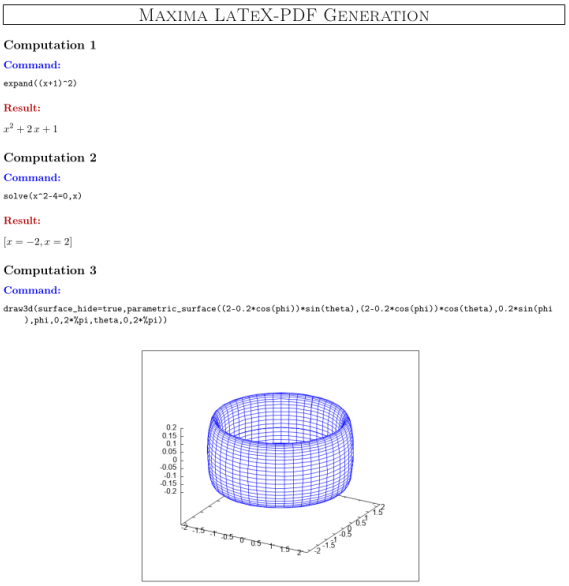
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Overview

This software is a utility that allows you to enter Maxima commands, either by typing them directly, by copying and pasting them, or by importing a `.mac` file containing Maxima commands. It generates a PDF file containing the entered commands and the results calculated by Maxima. It also allows you to export the LaTeX file used to generate the PDF, together with the document images, to a folder of your choice.

Screenshots





How the software works

The software sends commands to the Maxima computer algebra system (each command must end with either `;` or `$`). Maxima executes them and returns the results in LaTeX format. The software retrieves these data and lays them out in a LaTeX document, which is then compiled using `pdflatex`. It is a Python script, also available as an executable for Windows or Linux.

Comparison with wxMaxima

This utility is a lightweight, fast tool for quickly producing a high-quality document containing the results calculated by Maxima from user-entered commands, in PDF or LaTeX format. It does not offer the full capabilities of wxMaxima and is not suitable for solving complex exercises. For example, it can be used to generate a document containing calculations performed by Maxima, which can then be incorporated into a larger document.

Requirements

The software requires:

1. Maxima.
2. Gnuplot.

3. A LaTeX distribution including the following packages: `xcolor`, `listings`, `datetime2`, `babel`, `geometry`, `fancyhdr`, `lastpage`, `graphicx`, `mathtools`, `amsmath`, `amsfonts`, `amssymb`.
4. If you wish to use the source version (the Python script), Python must also be installed on your computer.

Installation instructions for each operating system are provided below.

Installation on Windows

Step 1: Install Maxima and add it to the PATH

► Download and install Maxima (version 5.49.0):

`https://sourceforge.net/projects/maxima/files/Maxima-Windows/5.49.0-Windows/`

► Edit the environment variables:

1. Open Settings: Windows key + I.
2. Go to System → About.
3. Click Advanced system settings.
4. In the window that opens, click the Environment Variables button at the bottom.
5. Select PATH and click Edit.
6. Add the following two entries at the end of the list:
   `C:\maxima-5.49.01\bin`
   `C:\maxima-5.49.01\gnuplot\bin`
7. Click OK as many times as needed to close all the windows.

Note: Gnuplot is installed automatically along with Maxima.

Step 2: Install a LaTeX distribution

If LaTeX is already installed, check that the following packages are installed: `xcolor`, `listings`, `datetime2`, `babel`, `geometry`, `fancyhdr`, `lastpage`, `graphicx`, `mathtools`, `amsmath`, `amsfonts`, `amssymb`. Otherwise, download and install the TeX Live distribution (it automatically installs all the required packages and updates the PATH):

`https://tug.org/texlive/windows.html`

Optional step 3: Install Python

If you wish to use the Python script rather than the executable, you need to install Python (the installation also adds it to the PATH automatically).

`https://www.python.org/downloads/windows/`

Step 4: Download mac2texpdf3.exe

Download the Windows version from:

`https://maxima-french-doc.fr/interfaces/`

Unzip the downloaded file, then double-click the extracted Windows executable (named `mac2texpdf3-w.exe`). Windows may display a warning indicating that the software is not certified. To run it, you need to allow the unknown application to run.

Python alternative: download the mac2texpdf3-w.py script

After unzipping the file, open a command prompt, navigate to the folder containing the Python file, and enter:

```
python3 mac2texpdf3-w.py
```

Installation on Linux

Step 1: Install Maxima

If Maxima is not installed, you can either install it using your distribution's package manager or download it from:

`https://sourceforge.net/projects/maxima/files/Maxima-Linux/5.49.0-Linux/`

The PATH is configured automatically, unlike on Windows.

Step 2: Install LaTeX

Use your distribution's package manager. Perform a full installation to obtain all the required packages. Otherwise, they are included in the `texlive-latex-extra` package.

Optional step 3: Install Python

If you wish to use the Python script rather than the binary, install Python using your distribution's package manager.

Step 4: Download mac2texpdf3.bin

Download the Linux version from:

`https://maxima-french-doc.fr/interfaces/`

Unzip the downloaded file, then double-click the extracted Linux binary (named `mac2texpdf3.bin`).
Python alternative: download the mac2texpdf3.py script
After unzipping the file, open a terminal, navigate to the folder containing the Python file, and enter:

```
python3 mac2texpdf3.py
```

Using mac2texpdf3

The software is very simple to use:

- Enter a series of Maxima commands in the text area. *Each command must end with `;` or `$`.*
- Choose the font size for the generated PDF. If there are plots, choose their scale (between 0 and 1).
- Click Generate PDF.
- Then click View PDF.
- To save the PDF and the corresponding LaTeX file, together with the document images, click the Export folder button.

A point to note

Since mac2texpdf3 does not number Maxima inputs and outputs, you cannot use `%n` to refer to the result of Maxima calculation number n. However, `%` is recognized and refers to the result of the previous calculation.

What's new in version 3.0 compared with version 2.0

Maxima input

- **Previous result `%`**: `%` now refers to the result of the previous command (for example, `(x+1)^2;` followed by `expand(%);`). Constants such as `%pi`, `%e` and `%i`, and strings containing `%`, are left unchanged.
- **Multi-line commands**: a command can span several lines. It ends at the first `;` or `$` outside a string or a comment. Several commands on the same line are also accepted.
- **Comments**: `/* ... */` comments, including nested and multi-line comments, are ignored. Lines starting with `*`, `%%` or `#` are also treated as comments.
- **`$` terminator**: commands ending with `$` are accepted. The `AFFICHER_RESULTAT_DOLLAR` option controls whether their result is displayed (`SHOW_RESULT_FOR_DOLLAR` in the English version).
- **Importing `.mac` files**: a new **Import .mac** button loads a Maxima file into the editor. UTF-8, cp1252 and Latin-1 encodings are detected.
- **Select all**: `Ctrl+A` (`Cmd+A` on macOS) works in the editor.

Graphics

- wxMaxima commands are converted automatically: `wxdraw2d`, `wxdraw3d`, `wxdraw`, `wxplot2d`, `wxplot3d`, `wximplicit_plot` and `wxcontour_plot`.
- Support for `implicit_plot`, `contour_plot` and `draw` with several scenes.

PDF / LaTeX output

- **Colored labels**: *Command* in blue, *Result* in dark red and *Output* in green.
- **Output from `print`, `printf`, `disp`, `display`, `for`, `while`...**: this is now captured and displayed in an *Output* block (it was previously lost). The returned value is no longer repeated as a *Result*.
- **Maxima messages** (errors, for example): displayed in the *Output* block.
- **Function definitions** (`define`, `:=`): displayed as plain text, without the trailing `;`, and with normal spacing.
- **Long lists**: split into lines of 10 complete elements. This fixes a LaTeX compilation error involving elements such as `\frac{5\pi}{6}` produced by Maxima.
- Commands containing backslashes or quotation marks are displayed correctly.

Platforms and languages

- Linux and Windows versions, each available in French and English.
- On Windows, `maxima.bat` is used, and paths passed to Maxima use `/`.

Requirements

Maxima, gnuplot (for graphics), and a LaTeX distribution providing `pdflatex` must be installed and available in the `PATH`.

Known limitations

- Animation commands (for example, `wxanimate_draw` and `with_slider_draw`) are not supported, since the output is a PDF/LaTeX file.
- `%o1`, `%i1` and `%th(n)` are not supported; use `%` for the previous result.

Links and references

Maxima documentation

`https://maxima-french-doc.fr/documentation/`

mac2texpdf3 software page

`https://maxima-french-doc.fr/interfaces/`

mac2texpdf3 development repository

The software is developed in a GitHub repository:
`https://github.com/michelprog59/mac2texpdf2`

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To request improvements or report bugs, you can either use GitHub directly or email me at michel.gosse@free.fr.